

Unit1. Vocabulary

Term	Definition
Science (Ciencia)	The systematic study of the structure and behavior of the physical and natural world through observation and experiment.
Technology (Tecnología)	The application of scientific knowledge and practical skills to create solutions that meet human needs or solve practical problems.
Technoscience (Tecnociencia)	The interdisciplinary context where scientific research and technological development are closely linked and mutually dependent.
Scientific Method (Método Científico)	A rigorous, systematic methodology used to acquire objective knowledge about the universe.
Observation (Observación)	Detecting and examining natural phenomena attentively.
Hypothesis (Hipótesis)	Proposing a provisional, reasoned explanation for observed phenomena.
Experimentation (Experimentación)	Designing tests and measuring quantitative variables to validate or reject a hypothesis.
Conclusion (Conclusión)	Determining whether experimental data supports the hypothesis.
Knowledge / "To Know" (Saber / Conocimiento)	The theoretical understanding of laws, theorems, and scientific models.
Execution / "To Make" (Hacer / Crear)	The practical action or process of building products, machines, tools, and processes.
Technological Process / Project Method (Proceso Tecnológico / Método de Proyectos)	An ordered sequence of phases used to turn a practical problem into an effective technical solution.
Design Constraints (Condicionantes / Restricciones)	Limitations or requirements imposed on a project, such as budget, safety, materials, dimensions, or time limits.
Problem Description (Descripción del problema)	Precise formulation of the need or challenge that must be solved.
Information Research (Búsqueda de información)	The phase dedicated to exploring existing solutions, consulting literature, checking catalogs, or interviewing experts.
Brainstorming (Lluvia de ideas)	A group creativity technique designed to generate many unfiltered ideas without early judgment.
Mind Mapping (Mapa mental)	A visual diagram used to organize ideas, concepts, and relationships around a central theme.
Sketch / Freehand Drawing (Boceto)	A quick, initial, freehand drawing of an idea without exact measurements.
Bill of Materials (Lista de materiales)	A detailed table listing all raw materials, components, tools, and estimated costs needed.

Process Sheet (Hoja de proceso / Planificación)	A document outlining step-by-step tasks, assigned team members, estimated durations, and safety guidelines.
Prototype (Prototipo)	An early operational model built to test, refine, and validate a design concept.
Assembly (Montaje)	The process of fitting components together according to design specifications.
Testing (Pruebas / Ensayos)	Assessing whether the constructed object meets all initial requirements and operates safely.
Redesign (Rediseño / Modificación)	Making adjustments or improvements based on test failures or evaluation outcomes.